

torch.diagonal#

<https://pytorch.org/docs/stable/generated/torch.diagonal.html>

Description:

Returns a partial view of input with the its diagonal elements with respect to dim1 and dim2 appended as a dimension at the end of the shape. The argument offset controls which diagonal to consider: If offset = 0, it is the main diagonal. If offset > 0, it is above the main diagonal. If offset < 0, it is below the main diagonal.

Function Signature

```
torch.diagonal(input, offset=0, dim1=0, dim2=1) → Tensor#
```

Parameters

Code Examples

```
>>> a = torch.randn(3, 3)
>>> a
tensor([[ -1.0854,  1.1431, -0.1752],
        [ 0.8536, -0.0905,  0.0360],
        [ 0.6927, -0.3735, -0.4945]])

>>> torch.diagonal(a)
tensor([ -1.0854, -0.0905, -0.4945])

>>> torch.diagonal(a, 1)
tensor([ 1.1431,  0.0360])

>>> b = torch.randn(2, 5)
>>> b
tensor([[ -1.7948, -1.2731, -0.3181,  2.0200, -1.6745],
        [ 1.8262, -1.5049,  0.4114,  1.0704, -1.2607]])

>>> torch.diagonal(b, 1, 1, 0)
tensor([1.8262])

>>> x = torch.randn(2, 5, 4, 2)
>>> torch.diagonal(x, offset=-1, dim1=1, dim2=2)
tensor([[[ -1.2631,  0.3755, -1.5977, -1.8172],
         [-1.1065,  1.0401, -0.2235, -0.7938]],

        [[ -1.7325, -0.3081,  0.6166,  0.2335],
         [ 1.0500,  0.7336, -0.3836, -1.1015]])])
```

Notes & Warnings

NOTE To take a batch diagonal, pass in `dim1=-2`, `dim2=-1`.

Detailed Documentation

Documentation

Returns a partial view of input with the its diagonal elements with respect to dim1 and dim2 appended as a dimension at the end of the shape.

The argument offset controls which diagonal to consider:

If offset = 0, it is the main diagonal.

If offset > 0, it is above the main diagonal.

If offset < 0, it is below the main diagonal.

Applying `torch.diag_embed()` to the output of this function with the same arguments yields a diagonal matrix with the diagonal entries of the input. However, `torch.diag_embed()` has different default dimensions, so those need to be explicitly specified.

input (Tensor) – the input tensor. Must be at least 2-dimensional.

offset (int, optional) – which diagonal to consider. Default: 0 (main diagonal).

dim1 (int, optional) – first dimension with respect to which to take diagonal. Default: 0.

dim2 (int, optional) – second dimension with respect to which to take diagonal. Default: 1.

To take a batch diagonal, pass in dim1=-2, dim2=-1.

